University of Rochester
Warner School of Education

Designing a Minecraft World That Carries Identity and Community Knowledge

Mini-Unit Packet

Layered Genius: Designing for Joy, Identity, and Disciplinary Skills

Grade 4–5 Computer Science / Digital Literacy mini-unit centered on Minecraft Education, multimodal meaning-making, identity-safe design, debugging, and disciplinary writing

Student	Piter Zacari Garcia Bautista
Course	EDU498: Literacy Learning as Social Practice
Session Focus	Towards a Pursuit of Skills
Assignment	Disciplinary Literacy Mini-Unit Extension
Mini-Unit Focus	Minecraft world-building as identity, community, coding literacy, and disciplinary writing
Student Profile	Ethan, Student 5, Computer Science
Submission Date	June 2026

This revision extends the earlier mini-unit through the pursuit of Skills. It treats Minecraft Education as a disciplinary literacy space where students build, code, debug, document, explain, and revise a meaningful digital world.

Revision note: *New additions are shown in blue. This version adds backward-design alignment, a computer science disciplinary writing component, and connections to CSTA and New York State Computer Science and Digital Fluency expectations for grades 4–6.*

Mini-Unit Snapshot

Task: extend the mini-unit to include disciplinary literacy and skill development, particularly through computer science writing, debugging documentation, and standards-aligned computational thinking.

Grade / discipline: Grade 4–5 Computer Science / Digital Literacy

Student profile: Ethan, a Seneca / Haudenosaunee and mixed-identity student who enjoys coding, game design, robotics, collaboration, outdoor learning, and community storytelling.

Mini-unit focus: Students use Minecraft Education to plan and begin building a small digital place that communicates identity, community knowledge, belonging, or future possibility.

Essential question: How can code, design, and digital place-making communicate who we are, what matters to us, and what our communities know?

End goal / backward design target: By the end of the unit, students will share a Minecraft place and a short Developer Design Note explaining how specific build/code choices communicate meaning for an audience. Evidence includes the world or screenshot, plan, debug log, peer feedback, and written/oral explanation.

Unit context and purpose: This three-lesson unit is designed for a Grade 4–5 Computer Science / Digital Literacy setting using Minecraft Education. Students use digital world-building as a literacy practice: they plan, code, debug, label, explain, and share a digital place that carries meaning. The purpose is to help students see coding as authorship, identity, community knowledge, and audience-aware design.

The unit foregrounds Joy and Identity by giving students choice, visible progress, and a protected way to connect design with who they are or what they value. The extension strengthens Skills by making disciplinary language and writing explicit: students name the problem they are solving, document steps, describe a bug or revision, and explain how computational choices shape audience understanding. Intellect and Criticality remain in the brainstorm phase, but the plan now names initial questions students will use to think about digital representation and access.

Three-lesson Arc:

- **Lesson 1: Plan a Meaningful Place.** Students choose a place concept, sketch meaningful features, decompose the build into steps, and draft a design goal that names audience and purpose.
- **Lesson 2: Build, Code, and Debug.** Students use Minecraft / MakeCode to build, test, revise, and document one coded or repeated feature that helps the world communicate meaning.
- **Lesson 3: Share, Explain, and Reflect.** Students present the world, submit a Developer Design Note, explain design choices, receive feedback, and revise one feature for clarity, welcome, or accessibility.

Across the arc, the disciplinary thread stays consistent: students move from idea to plan, from plan to build/code, from code to debugging evidence, and from debugging evidence to a short design explanation. This makes computer science skill visible without separating it from identity, joy, and community meaning.

Source Trail and Rationale for the Pick

This work grows from my work toward "*Layered Genius: Designing for Joy and Identity*" and from Muhammad's Identity pursuit, which frames identity as self-definition. It also carries forward my joy commitment of "belonging before building" and my placement observations of debugging, collaborative work, peer teaching, and student-led problem framing. I also draw on Moje, Luke, Davies, and Street's literacy-and-identity lens, Skerrett's attention to repositioning students as capable contributors, and Moje's disciplinary literacy framework. These sources help name coding, debugging, design, and explanation as literacy practices. Minecraft Education/MakeCode and the CSTA/ISTE standards serve as practical anchors for sequencing, debugging, computational thinking, creative communication, collaboration, and multimodal digital design.

For this revision, I also use the CSTA K–12 Computer Science Standards and the New York State Computer Science and Digital Fluency Learning Standards for grades 4–6. These discipline guidelines help move the mini-unit from a strong identity-centered design activity into a more explicit computer science literacy task. The focus is not only that students build something meaningful, but that they can use disciplinary practices such as decomposition, sequencing, debugging, documentation, design explanation, and audience-aware communication.

Student Profile and Identity Safety

Ethan's profile matters because the lesson can recognize him as a coder, designer, storyteller, and community-minded problem solver. At the same time, the lesson should honor his identity without requiring him to explain, display, or represent his culture for the class. Instead, it offers identity-safe choice by allowing students to build from lived experience, imagination, community, interests, family, land, language, hobbies, or future hopes. Because Ethan's profile includes Seneca / Haudenosaunee heritage, culturally specific symbols or stories should be student-chosen and handled with care. Private disclosure is never required. Fictionalized, symbolic, or aspirational places are valid.

Identity design move: The lesson treats identity as a design resource, not a disclosure requirement. Students choose what counts as meaningful, and the teacher assesses how clearly design choices communicate meaning. This keeps the work aligned with our focus on identity while respecting privacy, agency, and student control.

Discipline Standards and Skills Alignment

CSTA alignment: The unit aligns with upper-elementary computer science practices by asking students to decompose a larger project into smaller steps, use sequences or repeated patterns, test and debug their work, work collaboratively, and communicate about the design and purpose of a computing artifact.

New York State grades 4–6 alignment: The unit also aligns with NYS Computer Science and Digital Fluency expectations by supporting computational thinking, digital design, impacts of computing, collaboration, and communication. Students create a digital artifact, explain how it works, revise based on feedback, and consider how digital spaces can include or exclude community stories.

Disciplinary literacy move: In this mini-unit, writing is not separate from computer science. Students write like beginning designers/developers: they state a design goal, name steps, describe a bug or revision, explain how a feature works, and connect technical choices to audience meaning.

Pursuits Alignment

Pursuit	How to Address it
Joy	Students experience choice, visible progress, peer support, debugging as discovery, and the satisfaction of building something shareable.
Identity	Students design a meaningful place and explain how specific features communicate belonging, community, memory, culture, or future possibility.
Skills	Students practice sequencing, decomposition, loops or repeated patterns, debugging, spatial reasoning, screenshot use, partner communication, and oral/written explanation. They also complete a Developer Design Note that documents their design goal, steps, bug/revision, and audience meaning.
Intellect	Students analyze how code, space, signs, paths, color, interaction, and audience work together to create meaning in a digital world. Initial brainstorm: students learn that code is not only functional; it can communicate ideas, guide users, preserve memory, and shape how a digital place is read.
Criticality	Students ask whose stories do/don't appear, how accessibility shapes belonging, and what respectful use means when cultural symbols or community knowledge are involved. Initial brainstorm: students question who gets represented in games and digital worlds, whose knowledge is missing, and how designers can avoid using culture as decoration.

Multimodal and Multiliteracies Design

Minecraft functions here as a text students author. They read digital space, sketch a plan, build visually, use procedural thinking, explain orally, label in writing, and revise through peer feedback.

Modes included: spatial design, visual design, coding/sequencing, oral explanation, written labels or reflection, screenshots, collaboration, and peer feedback.

Access moves: sketch-first or build-first choice; projected model; partner roles; sentence stems; optional oral explanation; screenshots as memory support; chunked directions; low-floor/high-ceiling challenge options.

Why this supports multiliteracies: Students are not only writing or answering questions; they are composing meaning across space, image, code, talk, labels, screenshots, and peer feedback. A path, sign, repeated pattern, gathering space, or coded sequence can all function as evidence of meaning-making.

Disciplinary literacy layer: Students read their world as designers by asking what an audience can infer from paths, signs, repeated patterns, coded movement, and points of interaction. They write as beginning developers by documenting design goals, build/code steps, bugs, revisions, and audience meaning. The multimodal product and the written note work together: the world shows the design, while the note explains the technical and cultural choices behind it.

Lesson 1 Plan

Duration: 60 minutes

Lesson title: Planning and Prototyping a Meaningful Minecraft Place

Student-friendly target: I can plan and begin building a Minecraft place that shows something important about identity or community, and I can explain how my design choices carry meaning.

Identity-safe teacher language: Your place can be real, imagined, symbolic, collaborative, or future-facing. You do not have to share anything private. The goal is to make thoughtful design choices and explain what those choices communicate.

Objectives:

1. Students will choose a meaningful place connected to identity, community, belonging, or future possibility.
2. Students will break the place into smaller build/code steps.
3. Students will begin one prototype and test or revise at least one feature.
4. Students will explain one design choice using meaning, audience, and purpose.
5. Students will draft the first part of a Developer Design Note by naming their design goal and first 3–5 build/code steps.

Materials: Minecraft Education, teacher model world, projector, planning sheet, pencils or markers, optional MakeCode reference cards, exit slip, and Developer Design Note template. If devices fail, students complete a paper-world storyboard and mark which steps would later be coded.

Mins	Activity	Purpose
0–7	Show a generic Minecraft structure and a meaningful place. Ask: "What makes a place meaningful, not just well built?"	Frames design as meaning-making.
7–15	Model a small place plan: entrance, sign, path, gathering spot, and one repeated pattern that could be built or coded.	Makes decomposition and design choices visible.
15–25	Students choose a real, imagined, symbolic, or future place and sketch 3–5 important features.	Builds identity-safe agency.
25–38	Students label 3–5 build/code steps and begin a prototype in Minecraft or on paper. They copy these steps into the Developer Design Note.	Moves from idea to manageable action and disciplinary documentation.
38–50	Students test one feature, debug one issue, or revise one design choice.	Treats mistakes as part of learning.
50–57	Partner share: "This place matters because..." and "One choice I made was..."	Centers oral literacy and audience.
57–60	Exit slip: "What does your place show? What is your next build/code step?"	Captures evidence and next steps.

Differentiation: Students may work alone or with roles such as builder, navigator, debugger, or explainer. Beginners may build one strong scene; advanced students may add loops, agent movement, signage, or an audience-centered accessibility feature.

Standards connection: CSTA 1B-AP-10, 1B-AP-11, 1B-AP-15, and 1B-AP-17; ISTE Innovative Designer, Computational Thinker, Creative Communicator, and Global Collaborator. NYS Computer Science and Digital Fluency grades 4–6 connections include computational thinking, digital design, collaboration, communication, and impacts of computing.

Lesson 2 Plan

Duration: 60 minutes

Lesson title: Building, Coding, and Debugging the Meaningful Place

Student-friendly target: I can build and revise my Minecraft place so that at least one coded or repeated feature helps communicate meaning.

Joy + identity bridge: Lesson 2 keeps coding joyful by treating debugging as experimentation rather than failure. Students revise features so their worlds become clearer, more welcoming, or more meaningful to an audience. This connects technical skill to identity because the code is not isolated from the story of the place; it helps the place communicate belonging, care, memory, or future possibility.

Objectives:

1. Students will build at least two meaningful features from their plan.
2. Students will use a sequence, loop, repeated structure, or agent-based action to create/improve the world.
3. Students will debug one issue by naming the problem, testing a change, and recording what changed.
4. Students will document the debug cycle in disciplinary language: expected result, actual result, change made, and what the revision helped communicate.

Mins	Activity	Purpose
0–8	Revisit planning sheets and choose one “must-build” feature and one “try-to-code” feature.	Focuses students before building.
8–15	Teacher models a short debug cycle: predict, run/test, notice, revise. Teacher also models how to write this in a debug log.	Makes debugging visible, normal, and writable.
15–42	Students build and code with partner roles. Teacher conferences using the debug log.	Develops skills and persistence.
42–52	Screenshot checkpoint: students capture or sketch one feature and label what it means. Students add one sentence explaining how the feature works.	Connects product to identity, audience, and disciplinary explanation.
52–60	Quick share: one success, one bug, one next step.	Builds community expertise.

Assessment evidence: debug log, teacher conference note, screenshot/sketch checkpoint, and oral explanation. Students can also submit a brief before/after note showing how one design choice changed during debugging. The strongest evidence is the connection between the technical revision and the meaning students want to communicate.

Disciplinary writing evidence: Students complete the middle section of the Developer Design Note: “My bug or design problem was...” “I expected...” “What happened was...” “I changed...” “This helped my world communicate...”

Teacher look-fors: During conferences, the teacher listens for students naming what they tried or changed, and how the feature supports meaning. That evidence can be small: a revised pathway, clearer sign, repeated pattern, debug note, or screenshot label that connects a technical decision to identity, community, or audience.

Lesson 3 Plan

Duration: 60 minutes

Lesson title: Sharing Worlds, Reading Design, and Reflecting on Representation

Student-friendly target: I can present my Minecraft place, explain how it carries identity or community knowledge, and respond respectfully to other students' designs.

Representation bridge: Before students share, the teacher names that digital worlds are designed from choices. Some choices make people feel welcomed and seen; other choices erase people, flatten culture, or make access harder. Students are asked to explain what they chose, not to defend or expose private identity.

Objectives:

1. Students will explain at least two design choices and how they carry meaning.
2. Students will give peer feedback using notice/wonder language.
3. Students will reflect on whose stories are visible and how designers can make worlds more welcoming.
4. Students will complete and share a short Developer Design Note using computer science language and audience-aware reflection.

Mins	Activity	Purpose
0–8	Students prepare a two-sentence explanation and choose one screenshot or live view to show.	Supports concise presentation.
8–35	Gallery walk or paired screen-share: students explain their worlds and collect one notice/wonder comment.	Centers oral, visual, and digital literacy.
35–48	Whole-group critical question: "Whose stories do digital worlds usually show, and whose could be missing?"	Makes criticality explicit.
48–56	Students revise one sentence, sign, pathway, or accessibility feature based on feedback. They also revise the final sentence of the Developer Design Note.	Turns feedback into action.
56–60	Exit reflection: "My world shows..." "A designer can make digital spaces more welcoming by..."	Captures final evidence.

Assessment evidence: final screenshot/live demonstration, peer feedback note, short presentation, and exit reflection. Students can also submit a brief revision note showing how feedback changed one sentence, sign, pathway, or accessibility feature. The strongest evidence is the connection between the final design choice and how the student wants the world to be read by others.

Developer Design Note prompt: "My design goal was..." "The build/code steps I used were..." "One bug or revision I worked through was..." "One feature I want my audience to notice is..." "This world communicates identity, community, belonging, or future possibility by..."

Teacher look-fors: During sharing, the teacher listens for students explaining what their design shows, why specific choices matter, and how they responded to peer feedback. Evidence may include a clearer explanation, revised sign, accessibility feature, peer notice/wonder note, or reflection on welcome.

Assessment and Evidence

Evidence collected across the unit: planning sheet, debug log, screenshots or sketches, teacher conference notes, peer feedback, short presentation, Developer Design Note, and exit reflection.

Criterion	Meeting	Developing
Meaningful design	World includes features connected to identity, community, belonging, or future possibility.	World has features, but the meaning needs clearer explanation.
Computational thinking	Student decomposes, sequences, tests, debugs, or uses a repeated/coded feature.	Student builds but needs support naming steps or debugging.
Disciplinary writing	Student documents design goal, steps, bug/revision, and audience meaning using beginning computer science language.	Student explains the project generally but needs support connecting technical steps to meaning.
Multimodal literacy	Student uses building, visuals, code/planning, oral explanation, and written labels/reflection.	Student uses some modes but relies on one form of expression.
Critical awareness	Student can name whose stories are represented and how a design can welcome an audience.	Student notices audience or representation with prompting.

How the evidence is read: The teacher does not grade cultural disclosure or artistic polish. The main evidence is whether students can connect a design choice to a purpose, use basic computer science language to describe process, and revise after testing or feedback. This protects identity while keeping the task rigorous.

Sample and Appendix Tools

Teacher model sample: a small community garden world with an entrance path, welcome sign, gathering circle, repeated fence pattern, and one coded/agent-assisted repeated feature. The model should be meaningful but not perfect so students see revision as normal.

Planning card	Debug log	Developer note	Share card
• My place is: • This place matters because: • Three important features are: • My first 3–5 build/code steps are:	• I expected: • What happened: • I changed: • I learned:	• My design goal was: • My code/build steps were: • One bug or revision was: • My feature works by:	• My world shows: • One design choice I made was: • One way this world welcomes people is: • One question I still have is:

Student-facing routine: Students can use one card at a time instead of managing the whole packet at once: plan first, debug next, then complete the developer note and share card. This keeps the workflow accessible for students who need memory supports while still preserving a full evidence trail.

Reflection

This mini-unit shows how digital literacy can become a space for belonging. Joy is built into choice, creation, debugging, peer support, and visible progress. Identity is built into meaningful place-making and student control over what is shared. The lesson also protects privacy, which matters because identity-centered teaching should allow students to communicate meaning without forcing them to disclose trauma, culture, family history, or community knowledge on demand.

Muhammad's identity pursuit helps frame the work as self-definition. Moje, Luke, Davies, and Street help ask what identity the task makes possible. Skerrett supports the idea that students can be repositioned as capable contributors. Moje's disciplinary literacy frame helps name coding, debugging, design, and explanation as literacy practices.

The revision strengthens the mini-unit by making computer science writing visible. Students are not only building in Minecraft; they are learning how designers and beginning developers communicate a design goal, document a process, test and debug, respond to feedback, and explain how technical choices affect an audience. This keeps the unit aligned with disciplinary literacy because students practice the language and habits of computer science while still connecting skills to identity, joy, intellect, and criticality.

For Ethan specifically, the design matters because it does not ask him to simplify himself into one identity label. It gives him a way to connect coding, game design, land, community, mixed identity, and cultural knowledge only to the extent he chooses. That makes the technical work more humane: code becomes a tool for meaning-making, not a gatekeeping measure.

For my own teaching, the most important design decision is that students are assessed on how intentionally they communicate meaning. That distinction matters for neurodivergent, culturally connected, multilingual, and mixed-identity students because it keeps literacy tied to agency. In this unit, joy is not separate from rigor: students still decompose, debug, revise, document, and explain, but those skills serve a world they care about.

Appendix: Working-Memory-Friendly Lesson Flow Figures

Design note: These visual figures are meant to support teachers and students who benefit from seeing the whole learning path at once. Each figure separates the build sequence, teacher support, and evidence so the lesson can be followed without holding every step in working memory.

Teacher support: Model decomposition, protect private identity choices, offer roles, and show the first build step.

Evidence: Planning card, prototype or sketch, partner explanation, and exit slip.

Figure 1: Lesson 1 visual flow: Plan a Meaningful Place.

Teacher support: Normalize debugging, conference with short prompts, and connect technical changes back to meaning.

Evidence: Debug log, before/after note, screenshot label, and oral explanation.

Figure 2: Lesson 2 visual flow: Build, Code, and Debug.

Teacher support: Provide sentence stems, protect choice, guide honest feedback, and invite students to revise a small piece.

Evidence: Final screenshot or live view, peer feedback note, revision note, and exit reflection.

Figure 3: Lesson 3 visual flow: Share, Revise, and Reflect.